

TROC24_4sz_4ty operator list

James C Brakefield © 2025

register zero used to hold return address		five bit immediate field is value of -4..7, length - 1 of postfixied immediate or first 4 bits of 12 bit signed value				
register zero as index register reads as zero		last four registers are residue, frame pointer, stack pointer & PC				
Y scale factor is data size times (1 to 4)		each register has a 4-bit type code: unsigned, signed, float & Posit (28-bits total)				
format name	eventual layout (little endian)	usage	address arithmetic	instruction fields		
fmt16di	nnnnn dddd xxxx11	basic 16-bit immediate instruction N op D ->D ADD,SUB,MUL,DIV,CMP,AND,OR,XOR			8	
fmt16di	nnnnn dddd xxxx11	basic 16-bit immediate instruction N op D ->D IN,OUT,SHFT,FUNCT,LDI,LDIS,LDIF,LDITY			8	
	NNNNNNNN NNNNNNNN nnnnn dddd xxxx11	n-N instruction pairs: IN-LD, OUT-ST, SHFT-INSRT, FUNCT-EXTRCT		2nd op if n="10000"	4	
fmt16ds	sssss dddd xxxx01	basic 16-bit instruction S op D ->D ADD,SUB,MUL,DIV,CMP,AND,OR,XOR, MOV			9	
fmt16dn	nnnnn dddd xxxx11	load/store tagged at N: LDTN, STTN		N		
fmt16dn	nnnnn dddd xxxx01	AOB, SOB, BSRN, BRN, BRP, BRZ, BRNZ		PC + N	7	
fmt16dn	nnnnn dddd xxxx11	32 unary functions, small n		funcN(D)->D		
fmt24drsi	rrrrr ix sssss dddd xxxx10	basic 24-bit instruction		a type code		
fmt24drs	rrrrr 0xx sssss dddd xxxx10	R op S ->D		b base address register #		
fmt24dri	nnnnn 1xx sssss dddd xxxx10	immediate op S->D		c condition code predicate (execute instruction if condition true)		
fmt24dbx	jjjjj 0xx bbbbb dddd xxxx10	load/store base + scaled index	B + J * ds	d result register #		
fmt24dbn	nnnnn 1xx bbbbb dddd xxxx10	load/store base + offset	B + N	ds data size in bytes		
fmt24dr	kkkkk zxx sssss dddd xxxx10	op K -> D; 32 single operand instructions		e 2nd result register #		
fmt24dn	nnnnn nxx nnnnn dddd xxxx10	eleven bit signed offset or displacement	PC + N	i s is immediate field if i = 1		
				m immediate format: -4..4 or 12-bit or 1..3 byte immediate, useful for 12-bit immediate:		
fmt32drst	ttttt zzu rrrrr ix sssss dddd xxxx00	basic 32-bit instruction		n small constant or length of following immediate		
fmt32dbjs	rrrrr yyu jjjjj 0xx bbbbb dddd xxxx00	load from base + index * scaling + offset	B + J * Y + R	t 3rd operand register #		
fmt32dbjn	nnnnn yyu jjjjj 1xx bbbbb dddd xxxx00	load from base + index * scaling + offset	B + J * Y + N	u update residue enable, for ST instructions is store immediate enable		
fmt32dbjs	rrrrr yy0 jjjjj 0xx bbbbb dddd xxxx00	store to base + index * scaling + offset	B + J * Y + R	v swap operands enable		
fmt32dbjn	nnnnn yy0 jjjjj 1xx bbbbb dddd xxxx00	store to base + scaled index + offset	B + J * Y + N	w 4th operand register #		
fmt32mbjs	jjjjj yyl sssss 0xx bbbbb dddd xxxx00	store immediate to base + index * scaling + offset	B + J * Y + S	x op-code bits		
fmt32mbjn	nnnnn yyl jjjjj 1xx bbbbb dddd xxxx00	store immediate to base + index * scaling + offset	B + J * Y + N	y 2-bit code for: ds * (1..4)		
fmt32dr	cccc kku kkkkk zxx sssss dddd xxxx00	op-kkkkkkk (S) -> D; 128 single operand instructions		z negate/invert enables for one or two operands		
fmt32drst	rrrrr zzu ttttt ix sssss dddd xxxx00	op(R, S, T) -> D; triple operand instructions		k single operand function code		
fmt32dn	nnnnnnnnn nxx nnnnn nnnnn dddd xxxx00	nineteen bit signed offset or immediate	PC + N			
	40-bit instructions within 24-bit instruction code space, 9 op-code bits					
fmt40dsrct	ttttt xxx ccccc uxx rrrrr ix sssss dddd 11xx10	40-bit instruction with three source registers & predication				
fmt40dsret	ttttt xxx eeeee uxx rrrrr ix sssss dddd 11xx10	40-bit instruction with three source registers, 2 destination registers				
	48-bit instructions within 32-bit instruction code space, 12 op-code bits					
fmt48dsrct	ttttt eeeee xxxxxx ccccc uxx rrrrr ix sssss dddd 11xx00	basic 48-bit instruction with 3 source, 2 destination & predication				
fmt48dsrwt	ttttt eeeee xxxxxx wwwww uxx rrrrr ix sssss dddd 11xx00	basic 48-bit instruction with 4 source, 2 destination				
	Some 32-bit instructions support three operand operations					
	40-bit load/store instructions support predication					
	40-bit instructions support three operands and predication					
	40 and 48-bit instructions support explicit residue mechanism: register field(s) for extended operands and/or extended results					